Preface

Welcome to *Research Methods, Design, and Analysis*. You are embarking on a study that will help you to think systematically, critically, and creatively in Psychology and other disciplines. We have two primary goals for this text. First, we have focused on writing a book that provides an understanding of the research methods used to investigate human thought and behavior. Research methods tend to change slowly, but they do change. This book provides coverage of the complete range of research methods available today. Psychology tends to favor experimental methods, so we devote more time to experimental research methods. Because nonexperimental research also is used in many areas of psychology, we carefully cover this method, including how to write a proper questionnaire. Because of the rapid growth of qualitative and mixed methods in psychology, we carefully cover these methods to complement the more traditional methods and to add to each student's repertoire of research skills.

A second overarching goal that has been maintained throughout all editions of the textbook is to present information in a way that is understandable to students. We have attempted to meet this goal by presenting material in as simple and straightforward a manner as possible and by accompanying complex material with illustrations taken from the research literature. We believe that such illustrations not only assist in clarifying the presented material, but also bring the material to life when it is placed in the context of actual research studies. This allows the student not only to learn the material, but also to see how it is used in a research study.

Overview and Organization of the Textbook

Research Methods, Design, and Analysis is written at the undergraduate level and is intended for use in the undergraduate methods course. The book provides an introduction to all aspects of research methodology, and assumes no prior knowledge. The chapters are divided into seven major parts, as follows:

Part I. Introduction (Chapters 1 and 2)

This section begins with a discussion of knowledge and science in an effort to provide students with an understanding of the nature, goals, and outcomes of science. We believe that most students have an incomplete understanding of science and that they must understand its goals and limitations in order to appreciate and understand the nature of the research process. This is followed by a discussion of the major types of research used to investigate mind and behavior in an attempt to make sure that the students connect the various research approaches with science. We also discuss the major methods of data collection to help students see how empirical data are obtained.

Part II. Planning the Research Study (Chapters 3 and 4)

In this section, the focus of the book moves to some general topics involved in all research studies. First, we explain how to come up with a research idea, conduct a literature review, and develop a research question and hypothesis. Second, we explain the key ethical issues that must be considered when planning and conducting a research study. We explain the ethical guidelines sanctioned by the American Psychological Association.

Part III. Foundations of Research (Chapters 5–8)

In Part III, we cover some concepts that the researcher must understand before critiquing or conducting a research study. We begin with a discussion of measurement. We define measurement, and explain how measurement reliability and validity are obtained. Next, we explain how researchers obtain samples of research participants from targeted populations. We explain the different methods of random and nonrandom sampling, and we show the important distinction between random selection and random assignment. We also briefly explain the sampling methods used in qualitative research. Next we explain how research validity (i.e., valid results) is obtained. This includes discussions of the major kinds of research validity (internal, external, statistical conclusion, and construct) that must be addressed and maximized in empirical research. This section also includes a chapter explaining the control techniques required to obtain valid research results and a chapter explaining the procedure and details of carrying out a research study.

Part IV. Experimental Methods (Chapters 9–11)

Part IV is focused on, perhaps, the most prominent approach to research in psychology and related disciplines (i.e., experimental research). The section includes a chapter explaining how to select and/or construct a strong experimental research design, including the importance of factorial designs and coverage of main effects and interaction effects. Next, is a chapter on quasi-experimental designs including the nonequivalent comparison group design, the interrupted time-series design, and the regression discontinuity design.

The final chapter of this section covers single-case designs including ABA, ABAB, combination, multiple-baseline, and changing criterion designs.

Part V. Nonexperimental, Qualitative, and Mixed Methods Research (Chapters 12–14)

This section includes chapters on additional major research methods used in psychology and related disciplines. First, the student is introduced to the goals, design, and conduct of nonexperimental quantitative research. The student will also learn how to correctly construct a questionnaire and/or interview protocol. Second, the book includes a full chapter on qualitative and mixed methods research. The relative strengths and weaknesses of quantitative, qualitative, and mixed methods research are discussed, the different qualitative and mixed methods approaches and designs are explained, and information is provided about how to conduct a defensible and rigorous qualitative or mixed methods study.

Part VI. Analyzing and Interpreting Data (Chapters 15 and 16)

This section explains descriptive and inferential statistics in a way that is both rigorous and fully accessible to students with no prior background in statistics. The descriptive statistics chapter explains the graphic representation of data, measures of central tendency, measures of variability, measures of relationship between variables, and effect size indicators. Chapter 16 explains how researchers obtain estimates of population characteristics based on sample data and how researchers conduct statistical hypothesis testing. In an effort to connect design and analysis, the appropriate statistical tests for the experimental and quasi-experimental research designs covered in earlier chapters are discussed. The student will also learn how to present the results of significance tests using APA style.

Part VII. Writing the Research Report (Chapter 17)

In Part VII we explain the basics of writing a professional, informative, and accurate research manuscript that can be submitted for publication. The guidelines from the latest edition of the *Publication Manual of the American Psychological Association* are explained in this chapter.

Pedagogical Features

The pedagogical features include concept maps and objectives at the beginning of each chapter. Each chapter highlights important terms and concepts and includes definitions of these in the chapter margins. These terms and concepts are highlighted not only to point out to students that they are important, but also to increase the ease with which students can learn these terms and concepts. Study questions are spaced throughout each chapter to help students review the material after they have finished reading a section; this feedback system will assist students in learning the material and assessing whether they understand the material. Each chapter ends with several learning aids. First, a summary of the material, a list of the key terms, and a set of useful Internet sites are provided. Next, to help students assess their knowledge of the chapter material, a Practice Test is provided at the end of each chapter. These tests include several multiple choice questions that students can use to assess their knowledge of the chapter material. The Practice Test is followed by a set of Challenge Exercises; these are designed to provide students with exposure to and experiences with activities required in the conduct of a research study.

New to the Thirteenth Edition

Many minor changes have been made to the thirteenth edition to update references, clarify material, and improve the student learning process. The major changes are as follows:

1. All chapters and references were updated to reflect the current state of methods in psychology.
2. All chapters were carefully edited to increase clarity without sacrifice of rigor.
3. Chapter objectives are now directly connected to the first-level chapter headings.
4. Chapter 1. Historical coverage of science was slightly reduced. A new section on Science in the Twenty-First Century was added that stresses the empirical nature of psychological science and the concept of producing evidence (not proof) of scientific claims. Two additional characteristics of scientific research were added: Empirical, and Evidence Not Proof. Definitions of randomized controlled trials (RCTs) and operationalizing a construct were added. A new table on how to evaluate theories was added.
5. Chapter 2. Coverage of path analysis, natural manipulation research, and cross-sectional and longitudinal studies was removed because the book now has a full chapter on nonexperimental quantitative research A brief introduction to mixed methods research was added. A new section on advantages and disadvantages of nonexperimental quantitative research was added. Updated definitions were added for experimental research, manipulation, descriptive research, nonexperimental research, qualitative research, mixed methods research, field experiment, Internet experiment, correlational research, quantitatively driven mixed methods research, qualitatively driven mixed methods research, and equal-status or integrative mixed methods research.
6. Chapter 3. There is an increased focus on the central role of research questions. The section Bias in Research Ideas was revised and renamed Breadth of Research Ideas.

The goal is to focus on the positive side of the issue and recognize the need for diversity of questions and ideas. There is a brief addition explaining the parts of a research article. The section on evaluating web pages was reduced and coverage of Internet search engines was reduced.

7. Chapter 4. The section on dispensing with informed consent was deleted. The example IRB proposal was deleted, and instead, there is an updated discussion on submitting IRB proposals. Definition of anonymity was clarified.
8. Chapter 6. Coverage of experimenter expectancies and participant effects was simplified. A brief discussion of single and multiple-group designs was added to enhance the discussion of threats to internal validity. The discussion of accessible populations was deleted. The term "experimenter effects" was modified to "researcher effects" to increase the applicability of the concept. The same was done for "experimenter expectancies" (changed to "researcher expectancies") and "experimenter attributes" (changed to "researcher attributes"). A new table was added, titled Summary of Threats to Internal Validity for Single and Multiple-Group Designs.
9. Chapter 7 on control techniques was revised so that it is applicable for both experimental and nonexperimental research. Control techniques were reorganized, and statistical control was added. Coverage of researcher and participant effects was reduced, and instead, the focus is on techniques to reduce researcher and participant effects. New or revised/clarified definitions were provided for carryover effect, single-blind technique, partial-blind technique. Double-blind placebo method is now called the double-blind technique.
10. Chapter 8 was previously Chapter 9. It was edited so that this chapter is now applicable for both experimental and nonexperimental research. We added "type of statistical test" as an additional factor affecting statistical power.
11. Chapter 9. A brief addition was made to the section on factorial designs to indicate (a) at least one of the IVs (but not all) must be manipulated, and (b) factorials may (or may not) include pretests. These changes match the inferential statistics chapter. A marginal definition of rival hypothesis stating that it is a synonym for alternative explanation was added.
12. Chapter 10. A new table titled Summary of Quasi-Experimental Designs was added.
13. Chapter 11. A new table titled Summary of Single-Case Designs was added. Removed definition of withdrawal. Renamed the interaction design the *combination design* for clarity.
14. Chapter 12 is a new chapter. It includes the strengths and limitations of nonexperimental quantitative research (NQR). NQR designs are discussed, including cross-sectional and longitudinal designs and descriptive, explanatory, and predictive nonexperimental quantitative research designs.
15. Chapter 13. Two additional kinds of scales were added to Exhibit 13.1 (agreement scale and evaluation scale).
16. Chapter 14. A new table titled Summary of Qualitative Methods was added. Interpretive validity is now called interpretive or *emic* validity. Idiographic causation is now idiographic or *local* causation. Inside-outside validity in mixed methods research is now called emic–etic validity. Added two types of mixed methods validity: pragmatic validity and multiple-stakeholders validity.
17. Chapter 16. Coverage of nonparametric statistics was added, including a table titled Popular Nonparametric Statistical Procedures and Their Parametric Alternatives. The section titled Chi-Square for Contingency Tables is now Chi-Square Test for Independence. New marginal definitions of parametric statistics and nonparametric statistics were added.

Acknowledgments

As with all previous editions, we offer our sincere appreciation and gratitude to our students. They inspire and motivate us to create a book that respects their intelligence and invites them to be full participants in the research enterprise. We also thank all of our external reviewers of the present and past editions of this book. Last, we thank the Pearson production team.

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